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Size	
Submission	☒ NDA ☐ Renewal ☐ Variation change detail no.: Deficiency letter by NPRA dated 25-09-2017
Code of previous version	(44,45)-I-MY-04.05
Changes	Requested by NPRA to : change product name to: oral lyophilisate for sublingual xx mcg
	☐ CCDS version: ☐ SPC country/version/date: ☐ Core PIL version: ☐ LAC no.:
Name & Date	JJWO, 25-Sept-2017

MINIRIN®

Oral Lyophilisates [for sublingual](#) 60 mcg and 120 mcg

QUALITATIVE AND QUANTITATIVE COMPOSITION

MINIRIN® Oral Lyophilisate [for sublingual](#) 60 mcg:

One oral lyophilisate contains 60 mcg desmopressin (free base), as desmopressin acetate.

MINIRIN® Oral Lyophilisate [for sublingual](#) 120 mcg:

One oral lyophilisate contains 120 mcg desmopressin (free base), as desmopressin acetate.

Excipients: Gelatin (from fish), mannitol (E421) and citric acid, anhydrous (E330).

PHARMACEUTICAL FORM

Oral lyophilisate

MINIRIN® Oral Lyophilisate [for sublingual](#) 60 mcg:

White, round, oral lyophilisate marked with a drop shaped figure on one side.

MINIRIN® Oral Lyophilisate [for sublingual](#) 120 mcg:

White, round, oral lyophilisate marked with two drop shaped figures on one side.

THERAPEUTIC INDICATIONS

Central diabetes insipidus.

Primary nocturnal enuresis (from 5 years of age) when the enuresis alarm has not had the desired effect in patients with normal ability to concentrate urine.

Symptomatic treatment of nocturia in adults, associated with nocturnal polyuria, i.e. nocturnal urine production exceeding bladder capacity.

POSODOLOGY AND METHOD OF ADMINISTRATION

General

The dose of MINIRIN® Oral Lyophilisate is individually adapted. MINIRIN® Oral Lyophilisate 60 mcg is calculated to correspond to MINIRIN® tablet 0.1 mg, and 120 mcg corresponds to 0.2 mg (see also section Pharmacokinetic properties).

Desmopressin should always be taken at the same time in relation to food intake, since food intake causes decreased absorption and by that also might influence the effect of desmopressin, see section Interaction with other medicinal products and other forms of interaction.

In the event of signs of water retention/hyponatraemia (headache, nausea/vomiting, weight gain, and in serious cases convulsions) the treatment should be temporarily interrupted until the patient has completely recovered.

Indication specific

Central diabetes insipidus:

A suitable initial dose for children and adults is 60 mcg 3 times daily, administered sublingually. The dosage regimen is then adjusted in accordance with the patient's response. Clinical experience has shown, that the daily dose varies between 120 mcg and 720 mcg sublingually. For most patients, the maintenance dose is 60 mcg to 120 mcg sublingually 3 times daily. In the event of signs of water retention/hyponatraemia treatment should be temporarily interrupted and the dose should be adjusted.

Primary nocturnal enuresis:

A suitable initial dose is 120 mcg at bedtime, administered sublingually. The dose may be increased up to 240 mcg sublingually if the lower dose is not sufficiently effective. Fluid restriction shall be enforced. In the event of signs or symptoms of water retention and/or hyponatraemia the treatment should be interrupted until the patient has completely recovered. When the treatment is resumed strict fluid restriction is necessary, see section Special warnings and precautions for use. Evaluation of continued need of treatment should be carried out after three months by means of at least one treatment-free week.

Nocturia:

In nocturic patients, a frequency/volume chart should be used to diagnose nocturnal polyuria for at least 2 days and nights before starting treatment. A night-time urine production exceeding the functional bladder capacity or exceeding 1/3 of the 24-hour urine production is regarded as nocturnal polyuria.

The recommended initial dose is 60 mcg at bedtime, administered sublingually. If this dose is not sufficiently effective after one week, the dose may be increased up to 120 mcg sublingually and subsequently 240 mcg sublingually by weekly dose escalations. Fluid restriction should be enforced.

Treatment should not be initiated in the elderly (65 years of age and over). Should treatment of these patients be considered, serum sodium should be measured before beginning of treatment and after 3 days of treatment. The same applies at increase in dosage and other occasions during treatment as deemed necessary by the treating physician, see section Special warnings and precautions for use. In the event of signs or symptoms of water retention and/or hyponatraemia the treatment should be interrupted until the patient has completely recovered. When the treatment is resumed strict fluid restriction is necessary, see section Special warnings and precautions for use.

If adequate clinical effect is not achieved within 4 weeks following appropriate dose titration the treatment should be discontinued.

Special Populations

Paediatric Population: MINIRIN® Oral Lyophilisate is indicated in Central Diabetes Insipidus and Primary Nocturnal Enuresis (see section Pharmacodynamic properties and indication specific information above). Dose recommendations are the same as in adults.

Elderly: Treatment of nocturia should not be initiated in patients >65 years (see dosage section above).

Renal Impairment: see section Contraindications.

Hepatic Impairment: see section Interaction with other medicinal products and other forms of interaction.

Administration

MINIRIN® Oral Lyophilisate is placed under the tongue where it is dissolved without water.

CONTRAINDICATIONS

- Habitual or psychogenic polydipsia (urine production exceeding 40 ml/kg/24 hours);
- Known or suspected cardiac insufficiency and other conditions that require treatment with diuretics;
- Moderate and severe renal insufficiency (creatinine clearance below 50 ml/min);
- Syndrome of inappropriate ADH secretion (SIADH);
- Known hyponatraemia;
- Hypersensitivity to the active substances or to any of the excipients (contain fish gelatine).

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

In patients with urgency/urge incontinence, organic causes for increased micturition frequency or nocturia (e.g. benign prostate hyperplasia, urinary tract infection, bladder stone/tumor), polydipsia and poorly adjusted diabetes mellitus, the specific cause should be treated primarily.

At treatment of primary nocturnal enuresis and nocturia, fluid intake should be limited to the least possible during the period of 1 hour before evening dose until at least 8 hours after administration. Treatment without concomitant reduction in fluid intake may lead to water retention and/or hyponatraemia (headache, nausea/vomiting, weight gain and in serious cases convulsions).

All patients and, when applicable, their guardians should be carefully instructed to adhere to the fluid restrictions.

In clinical trials, higher occurrence of hyponatraemia was found in patients over 65 years. Therefore, treatment should not be initiated in the elderly, especially not in patients suffering from other conditions that may increase the likelihood of fluid or electrolyte imbalance.

Elderly patients, patients with low serum sodium levels and patients with a high 24-hour urine volumes (above 2.8 to 3 litres) have an increased risk for hyponatraemia.

Precautions to avoid hyponatraemia including careful attention to fluid restriction and more frequent monitoring of serum sodium must be taken at:

- concomitant treatment with drugs known to induce syndrome with disturbed ADH secretion (SIADH), e.g. tricyclic antidepressants, selective serotonin reuptake inhibitors (SSRI), chlorpromazine and carbamazepine,
- concomitant treatment with NSAIDs.

Precautions must be taken in patients at risk for increased intracranial pressure.

Treatment with desmopressin should be interrupted during acute illnesses characterised by fluid and/or electrolyte imbalance such as systemic infections, fever, gastroenteritis.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Substances that are known to induce disturbed ADH-secretion, e.g. tricyclic antidepressants, SSRI, chlorpromazine and carbamazepine as well as some antidiabetics of the sulfonylurea group, especially chlorpropamide, may cause an additive antidiuretic effect with an increased risk of fluid retention (see section Special warnings and precautions for use).

NSAID preparations may induce water retention/hyponatraemia (see section Special warnings and precautions for use).

Concomitant treatment with loperamide may result in a three-fold increase in desmopressin plasma concentration, which may lead to an increased risk of water retention and/or hyponatraemia.

Other drugs slowing intestinal transport might have the same effect. However, this has not been investigated.

Concomitant treatment with dimethicone may result in a decreased absorption of desmopressin.

It is unlikely that desmopressin interacts with pharmaceuticals affecting hepatic metabolism, since desmopressin has not been shown to undergo any significant liver metabolism in *in vitro* studies with human microsomes. However, no formal interaction studies *in vivo* have been carried out.

Concomitant food intake has not been investigated with MINIRIN® Oral Lyophilisate. A standardised meal with 27% fat taken together with or 1.5 h prior to MINIRIN® tablet decreased the extent and rate of absorption of desmopressin by 40%. No significant effect was observed with respect to pharmacodynamics (urine production or osmolality). It cannot be excluded that some patients may have decreased antidiuretic effect at concomitant food intake.

PREGNANCY AND LACTATION

Fertility

Fertility studies have not been carried out. *In vitro* analysis of human cotyledon models have shown that there is no transplacental transport of desmopressin when administered at therapeutic concentrations corresponding to recommended dose.

Pregnancy

Data on a limited number (n = 53) of exposed pregnancies in women with diabetes insipidus as well as data on a limited number (n = 54) of exposed pregnancies in women with von Willebrand disease indicate no adverse effects of desmopressin on pregnancy or on the health of the foetus/newborn child. No other relevant epidemiological data are available. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development.

Caution should be exercised when giving MINIRIN® Oral Lyophilisate to pregnant women.

Lactation

Results from analyses of milk from nursing mothers receiving high dose desmopressin (300 mcg intranasally), show that desmopressin is transferred to the milk but that the amount of desmopressin that can be transferred to the child is low and probably less than the amounts required to influence diuresis. Whether desmopressin will accumulate in breast milk upon repeated doses has not been studied.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

MINIRIN® has no or negligible effect on the ability to drive vehicles and use machines.

UNDESIRABLE EFFECTS

Summary of the safety profile

The most serious adverse reaction with desmopressin is hyponatraemia, see below under “Description of selected adverse reactions”.

In adults the most commonly reported adverse reaction during treatment was headache (12%). Other common adverse reactions were hyponatraemia (6%), dizziness (3%), hypertension (2%) and gastrointestinal disorders (<1%). Anaphylactic reactions have not been seen in clinical trials but spontaneous reports have been received.

In children the most commonly reported adverse reaction was headache (1%). Less common were psychiatric disorders (<1%), which generally abated after treatment discontinuation and gastrointestinal disorders (<10%). Anaphylactic reactions have not been seen in clinical trials but spontaneous reports have been received.

Tabulated summary of adverse reactions

Adults:

The frequency of adverse drug reactions reported in clinical trials with oral desmopressin conducted in adults for treatment of nocturia (N=1557) combined with the post marketing experience for all adult indications (incl. central diabetes insipidus). Reactions only seen post marketing have been added in the 'Not known'-frequency column.

MedDRA Organ Class	Very common (≥1/10)	Common (≥1/100, <1/10)	Uncommon (≥1/1000, <1/100)	Rare (≥1/10 000, <1/1000)	Not known (cannot be estimated from available data)
Immune system disorders					Anaphylactic reaction
Metabolism and nutrition disorders		Hyponatraemia*			Dehydration**, Hypernatraemia**
Psychiatric disorders			Insomnia	Confusional state*	
Nervous system disorders	Headache*	Dizziness*	Somnolence, Paraesthesia		Convulsions*, Asthenia**, Coma *
Eye disorders			Visual impairment		
Ear and labyrinth disorders			Vertigo*		
Cardiac disorders			Palpitations		
Vascular disorders		Hypertension	Orthostatic hypotension		
Respiratory, thoracic and mediastinal disorders			Dyspnoea		
Gastrointestinal disorders		Nausea*, Abdominal pain*, Diarrhoea, Constipation, Vomiting*	Dyspepsia, Flatulence, bloating and distension		
Skin and subcutaneous tissue disorders			Sweating, Pruritus, Rash, Urticaria	Dermatitis allergic	
Musculoskeletal and connective tissue disorders			Muscle spasms, Myalgia		
Renal and urinary disorders		Bladder and urethral symptoms			
General disorders and administration site conditions		Oedema, Fatigue	Malaise*, Chest pain Influenza like symptoms		
Investigations			Weight increased*, Hepatic		

			enzyme increased, Hypokalaemia		
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*Hyponatraemia may cause headache, abdominal pain, nausea, vomiting, weight increase, dizziness, confusion, malaise, vertigo and in serious cases convulsions and coma.

** Only seen in the CDI indication

Paediatric Population:

The frequency of adverse drug reactions reported in clinical trials with oral desmopressin conducted in children and adolescents for treatment of primary nocturnal enuresis (N = 1923). Events only seen in post marketing have been added in the 'Not known' frequency column'.

MedDRA Organ Class	Common (≥1/100, <1/10)	Uncommon (≥1/1000, <1/100)	Rare (≥1/10 000, <1/1000)	Not known (cannot be estimated from available data)
Immune system disorders				Anaphylactic reaction
Metabolism and nutrition disorders				Hyponatraemia ****
Psychiatric disorders		Affect lability**, Aggression***	Anxiety symptoms, Nightmare****, Mood swings****	Abnormal behaviour, Emotional disorder, Depression, Hallucination, Insomnia
Nervous system disorders	Headache*		Somnolence	Disturbance in attention, Psychomotor hyperactivity, Convulsions*
Vascular disorders			Hypertension	
Respiratory, thoracic and mediastinal disorders				Epistaxis
Gastrointestinal disorders		Abdominal pain*, Nausea*, Vomiting*, Diarrhoea		
Skin and subcutaneous tissue disorders				Dermatitis allergic, Rash, Sweating, Urticaria
Renal and urinary disorders		Bladder and urethral symptoms		
General disorders and administration site conditions		Oedema peripheral, Fatigue	Irritability	

*Hyponatraemia may cause headache, abdominal pain, nausea, vomiting and in serious cases convulsions

**Post marketing reported equally in children and adolescents (<18years)

***Post marketing almost exclusively reported in children and adolescents (<18years)

****Post marketing reported primarily in children (<12years)

Special populations:

Elderly patients and patients with low serum sodium levels may have an increased risk of developing hyponatraemia (see section Special warnings and precautions for use).

Description of selected adverse reactions:

The most serious adverse reaction with desmopressin is hyponatraemia, which may cause headache, abdominal pain, nausea, vomiting, weight increase, dizziness, confusion, malaise, vertigo and in serious cases convulsions and coma. The cause of the potential hyponatraemia is the anticipated antidiuretic effect. Hyponatraemia is reversible and in children it is often seen to occur in relation to changes in daily routines affecting fluid intake and/or perspiration. In adult study subjects treated for nocturia, the majority of those developing low serum sodium, developed this after 3 days of treatment or after dose increase.

Special precautions should be observed in adults as well as in children, see section Special warnings and precautions for use.

OVERDOSE

Toxicity

Overdosage leads to prolonged duration of action with an increased risk of fluid retention and hyponatraemia. Even normal doses may in combination with considerable fluid intake cause water intoxication. Doses from 0.3 mcg/kg i.v. and 2.4 mcg/kg intranasally have caused hyponatraemia and convulsions in children and adults. On the other hand, 40 mcg intranasally administered to a 5-month-old baby and 80 mcg intranasally to a 5-year-old gave no symptoms. 4 mcg administered parenterally to a newborn caused oliguria and weight gain.

Symptoms

The same symptoms as for water intoxication. Headache, nausea. Fluid retention, hyponatraemia, hyposmolality, oliguria, CNS depression, convulsions, pulmonary oedema. See also side-effects described in the text.

Treatment

The treatment of hyponatraemia should be individual but the following general recommendations may be given: Hyponatraemia is treated by interrupting the desmopressin treatment and fluid restriction. If the patient has symptoms an infusion of isotonic or hypertonic sodium chloride may be given. When the fluid retention is serious (convulsions and loss of consciousness) furosemide treatment is given.

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: vasopressin and analogues

ATC code: H01B A02.

MINIRIN® contains desmopressin, a structural analogue of the natural pituitary hormone arginine vasopressin. The difference lies in that the amino group in cysteine has been removed and L-arginine has been substituted by D-arginine. This results in a considerably longer duration of action and a complete lack of pressor effect in the dosages used clinically.

Clinical trials with desmopressin tablets in the treatment of nocturia showed the following:

- A reduction of at least 50% in the mean number of nocturnal voids was obtained in 39% of patients with desmopressin compared to 5% of patients with placebo ($p < 0.0001$).
- The mean number of voids per night decreased by 44% with desmopressin compared to 15% with placebo ($p < 0.0001$).
- The median duration of first undisturbed sleep period increased by 64% with desmopressin compared to 20% with placebo ($p < 0.0001$).
- The mean duration of first undisturbed sleep period increased by 2 hours with desmopressin compared to 31 minutes with placebo ($p < 0.0001$).

PHARMACOKINETIC PROPERTIES

Absorption

The plasma concentration of desmopressin has been shown to be equivalent after administration of MINIRIN® tablet 2x0.2 mg (0.4 mg) and MINIRIN® Oral Lyophilisate 240 mcg respectively.

The mean bioavailability of desmopressin administered sublingually as MINIRIN® Oral Lyophilisate is approximately 0.25%. Concomitant intake of food has not been investigated with MINIRIN® Oral Lyophilisate but concomitant food intake with MINIRIN® tablets decreases the rate and extent of absorption by 40%. Desmopressin exhibits a moderate to high variability in bioavailability, both within and between subjects. The plasma concentration of desmopressin increases proportionally to administered doses and after administration of 200, 400 and 800 mcg C_{max} was 14, 30 and 65 pg/mL respectively. T_{max} was reached after 0.5 – 2 hours. The terminal half-life is estimated to 2.8 hours.

Distribution

The distribution volume of desmopressin after intravenous administration is 33 L (0.41 L/kg). Desmopressin does not cross the blood-brain barrier.

Metabolism

In vitro studies with human liver microsomes have shown that no significant amount of desmopressin is metabolized in the liver. It is therefore unlikely that desmopressin is metabolized in the liver in human beings.

Elimination

52% of the amount of administered desmopressin is found in the urine.

Paediatric Population

The population pharmacokinetics has been studied in children (6-12 years) with PNE and is comparable to that in adults.

INCOMPATIBILITIES

Not applicable.

SHELF LIFE

3 years.

SPECIAL PRECAUTIONS FOR STORAGE

Do not store above 30°C.

Store in the original package in order to protect from moisture and light.

NATURE AND CONTENT OF CPACK SIZES

Aluminium/Aluminium blisters of 10 oral lyophilisates.

Pack sizes:

60 mcg: 30 oral lyophilisates.

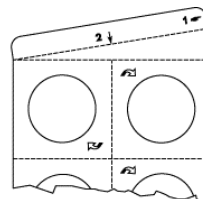
120 mcg: 30 oral lyophilisates.

Not all pack sizes may be marketed.

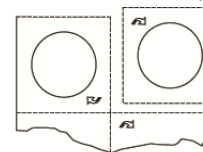
SPECIAL PRECAUTIONS FOR DISPOSAL AND OTHER HANDLING

The oral lyophilisates are brittle and could not be pressed through the foil since they may break. The oral lyophilisates should be taken out from the blisters by removing the aluminium cover.

1. Remove the end tab completely starting from the corner where a hand symbol is printed.



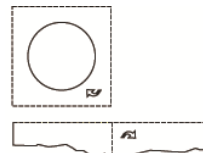
2. Tear off the blister unit along the vertical



3. Peel off the foil starting at the corner with the printed arrow. The product can then be removed from the pack.



4. To access the next tablet, tear off the blister unit along the horizontal perforation.



MANUFACTURER

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DATE OF REVISION

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